COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S4	**SECTION:**	CPE21S4
DATE PERFORMED:	18/09/2026	**DATE SUBMITTED:**	18//09/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a binary search tree?

- A binary search tree represents a specialized node-based data structure wherein each parent element possesses no more than two child nodes. Values less than the key are positioned within the left subtree, whereas greater values reside in the right subtree, thereby enabling efficient search operations.

2. Where can a binary search tree be used?

- Binary search trees are used anywhere you need to find or sort data quickly. You see them in phone contacts, dictionaries, spell checkers, and databases. They help keep things in order without extra work.

3. What is a tree traversal?

- A tree traversal is just visiting every single part of the tree one by one. You go through every item so you don’t miss anything. There are different ways you can do this.

4. Differentiate a post-order and pre-order traversal; give examples.

- Pre order traversal checks the current item first before moving to the left and right sides. For a tree with 5 at the top, 3 on the left, and 7 on the right, pre-order goes 5, 3, 7. Post-order checks both sides first and looks at the current item last. The same tree in post-order goes 3, 7, 5.

4. What is a parse tree and its use?

- A parse tree breaks down sentences or math formulas into smaller pieces so computers understand them clearly. It shows which parts belong together and what to do first. Programmers use it to read code correctly and calculate math problems in the right order.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **did not replace original thinking**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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